

Controller Design Report

DEN 161

Chase Harvey

C Harvey Group 7

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Jack Meirowsky, Ian Dodson, Kaiden O'Keefe, Max, Requet

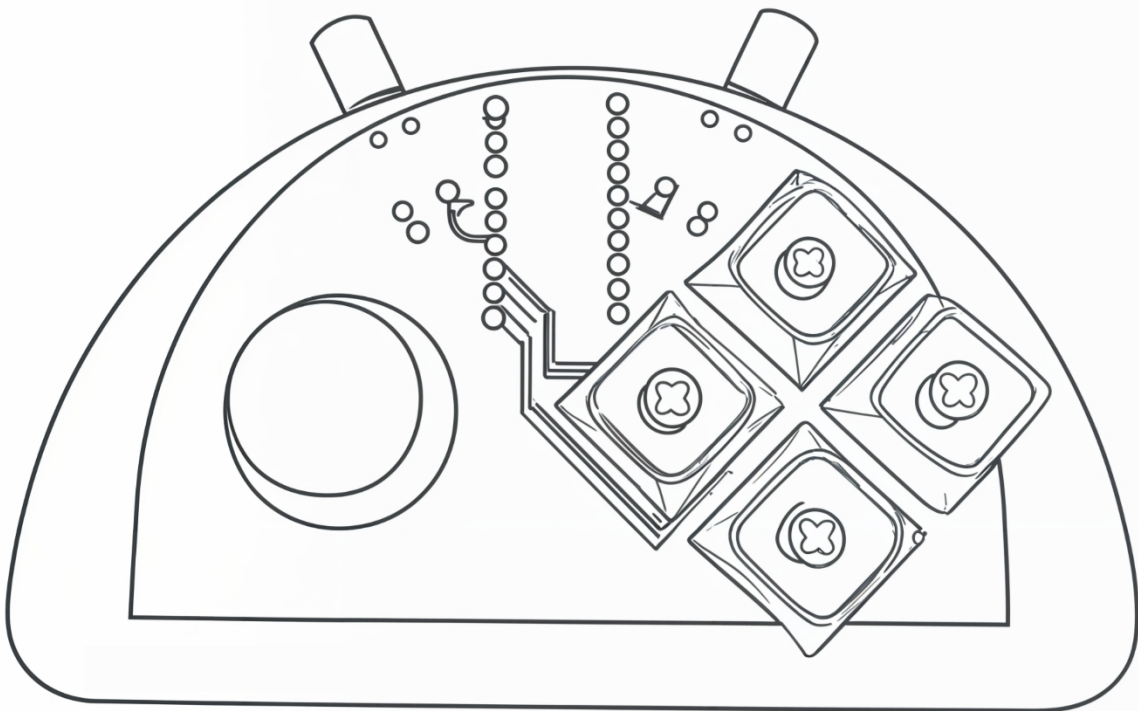


Figure 1 - Controller Design

Executive Summary

In this project, our group was tasked with designing a game controller and practicing every step of the engineering process. We were given a list of parameters that we had to meet as we tried to design the best product we could. These parameters included: safety, cost, size, compatibility, and comfortability. There were several solutions to this project and after brainstorming and discussing we decided on the design seen in Figure 1. Once we had our design, we worked to meet all the criteria and make the best controller possible. Throughout this semester long project, we practiced real world engineering skills like KiCad design, coding in the C+ language, collecting and analyzing data to improve our product, and dividing work up as a team. This course has given us a great sneak peek into the engineering process and taught us several valuable skills, such as designing a functioning product around a list of criteria and learning how to communicate and pitch our product.

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1. Problem Background

Our goal is to design a working game controller that can be used to play at least 3 different doodle games utilizing the game's full functionality. We are designing this controller for an abled body person of 16 years or older. The device must fit in a standard shipping box, cost under \$60, and be able to be put together and understood by the user. Some interested parties include our user and the K-State College of Engineering. The user wants to have good experience using the controller, so it needs to be comfortable and intuitive. The College of Engineering wants to see a design that meets all the criteria mentioned above, so our controller needs to meet all the limitations to the best of its ability.

Our controller has a good design right now for comfortability and intuition. The semi-circle shape allows for a natural hand grip while using the device, making it easy to use for long periods of time without hand pain. Our design also uses typical button mappings that are seen in a lot of modern controllers. The buttons make it easy to understand how the controller works and how to use it. However, our joystick is used to control the mouse with the bumpers for the left and right clicks. This may be a little strange at first, but once you get the hang of it, it should work great for the controller and allow it to do more. Overall, the construction and functionality of the controller should be simple if we include some form of instruction manual to resolve anything that needs clarification.

2. Design Criteria

Interested Party	Design Criteria (Want)	Criteria Description (Function)	Numerical Constraint (Metrics)
Employer	Safety	The controller is safe for an able bodied 16-year-old according to professionals	A scale of 1-10 (10 means the safest); average score is 8 or higher
Investor	Cost	The controller production cost fits within the budget	All of the components together cost less than \$60
Investor	Size	All of the components of the controller are able to fit into a standard shipping box	The width and height of all the pieces is less than 12" x 12"
User	Compatibility	The controller can access all the features on at least 3 google doodle games	Survey at least three people OUTSIDE your group. Have them use your controller to play three different google doodle games. Have them give a score using following scale: 5: Easily played three games. 4: Played three games but would have preferred to use a keyboard/mouse. 3: Only compatible with 2 games

			2: Only compatible with 1 game 1: Controller didn't work The average score must be a 4 or higher.
User	Comfort	The controller is comfortable to use and doesn't cause hand pain	On a scale of 1 – 10 (10 is most comfortable)

Table 1 - Design Criteria

Safety is essential to make sure that our controller doesn't harm the user in any way. The user's safety comes first when designing a product and it is extremely important to ensure that it operates appropriately and safely. The cost of the product is important aspect of the design of a product because you want to make sure that you can afford to make and produce it. If your product is too expensive, no one is going to buy it. The size of the product matters for advertising and distribution. If we built a massive controller, it would be unrealistic to use – the same goes for a tiny controller. The compatibility of the controller is ultimately the main aspect of it. It needs to be able to be used to play the games well to create a positive user experience. Comfort is another important criteria for the user. They want to have a comfortable time when they use our device; we want them to come back and use our controller again.

3. Evolution of Design

3.1. Design Selection

Board Shape	Comfort	Look	Compatibility	Total
Semi-circle	9	8	8	25
Rectangle	6	8	7	21
Triangle	4	7	6	17

Table 2 - Decision Matrix

For the comfort of the design, we considered how the controller would feel to hold. Because the semi-circle design is rounded, it is going to be more comfortable with the natural curve of the hand as you hold the device. This is seen in popular modern-day controllers that have a rounded look. The rectangle and triangle shapes are too ridged and would be uncomfortable to use. When looking at controllers in the real world, it is more popular to use a rounded controller nowadays - like X-box or PlayStation - than a rectangular one - like the original Nintendo controller. The look of the device comes down to the individual. Overall, the look of the semi-circle and rectangle concepts looked the most appealing while the triangle shape looks a little silly but unique. The compatibility has to do with the placement of our buttons and joysticks and how they complement the shape. Because of the roundness, your fingers are going to tend towards the middle where the joystick and buttons are. The rectangular shape doesn't really work with the joystick but would be fine. The triangle shape – similar to its comfortability – would be hard to find a good spot to put the elements. Based off the decision matrix, the semi-circle design works best for our controller.

3.2. Iterative Process

Our design is based off a semi-circle displayed in the picture on the title slide. This can be altered to our needs such as the need for a clicking button, or a second joystick are some further alterations that can be figured in the future, same with a home centralized button for a return feature. Once we have decided on our games, we will have a better idea of which features need to be implemented into our controller design.

4. Final Design Description

The semi-circle controller, included with a left joystick, 4 button diamond in all 4 directions, and the 2 bumpers. Follows an ergonomic flow to emphasize comfort along with efficiency in a simple yet complex manner. The 4 diamond buttons are coded for WASD for movement while playing games.

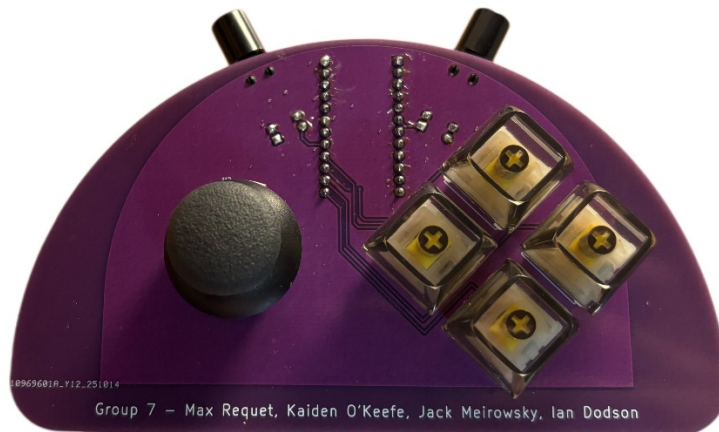


Figure 2 - Final Prototype

5. Prototype Analysis

To test our design, we had family members use our controller and score the product based on our criteria. The testers were given the controller and 3 Google Doodles games with no extra instructions. We then took their responses to adjust the product to make it more user friendly.

The testers were given “Snake”, “Minesweeper” and “Garden Gnomes” to play as we believed they best displayed the functionality of the controller.

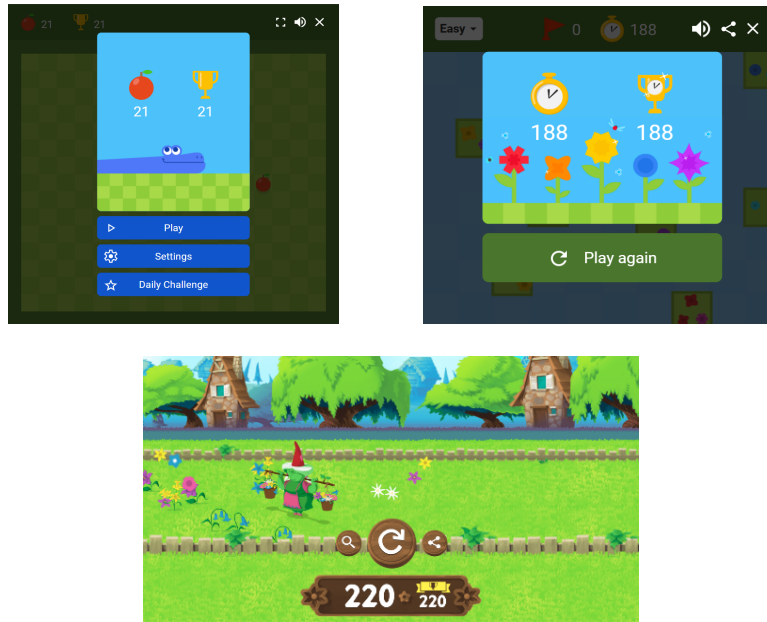


Figure 3 - Google Doodle Testing

User	Compatibility	Comfort	Comments
Family Member #1	6/10	5/10	Slow down cursor
Family Member #2	7/10	6/10	Change space button to joystick
Family Member #3	9/10	8/10	Add arrow key functionality to buttons

Table 3 - Family Testing Ratings

- To fix the cursor sensitivity, the output range of the mouse controls was lowered from 6 to 4, limiting the amount the mouse is able to move every cycle, slowing it down.
- To fix the space button and arrow keys, new `keyboard.press()` methods were added to allow the controller to use the requested keyboard inputs.

6. Verification Methods

6.1 Safety:

To assess safety, we meet with Dr. Marty Kump, a professor in the Department of Electrical and Computer Engineering at K-State. Dr. Kump has extensive experience with micro-controllers and electronics, making him a reliable safety consultant. Dr. Kump saw no safety issues with our controller and was very impressed with the overall design. He scored our controller 10/10 for safety.

6.2 Cost:

To calculate the cost, we created an R&D BOM for our prototype. Using our list of parts, we found the cost to purchase all the items and determined how much it would cost us to make one controller with the online sales. Some of the items came with more parts than we would need, but we included the price to add leeway and ensure our prototype was under the \$60 limit. As seen in the BOM, the cost to produce the one prototype controller was \$58.14, satisfying the criteria. We then calculated the cost per controller if we were to manufacture 100 controllers. The total cost per controller was \$16.49, showing just how much can be saved by producing it in bulk.

6.3 Size:

To satisfy these criteria, we made sure that our control could fit inside of the box provided to us by our instructor. Our prototype fit in it with ease and had a significant amount of space to spare.

6.4-6.5 Compatibility & Comfort:

For these two criteria, we gave our controller to other students and family members to get their opinions on how well the controller was able to play the game and how comfortable they felt playing it. We used the average ratings of all our users to determine our final score in each category.

User	Compatibility	Comfort
Student 1	8/10	6/10
Student 2	9/10	5/10
Student 3	7/10	6/10
Student 4	9/10	6/10

Table 4 - Student Testing Results

User	Compatibility	Comfort
Family Member 1	6/10	5/10
Family Member 2	7/10	6/10
Family Member 3	9/10	8/10

Table 5 - Family Testing Results

Average Compatibility – 7.9/10

Average Comfort – 6/10

7. Data Collection

The data was collected by asking members of other groups in our class and family members to rank our controller on a scale of 1-10 for compatibility and comfort. Compatibility was evaluated by how many of the features the control could access as well as how well the controller could play the Google Doodle games receiving an average of 7.9/10. Comfort was evaluated on how naturally the controller sat in the users' hands and how easy it was to move the joysticks and push the buttons. Comfort received an average of 6/10.

8. Design Criteria Analysis (Verification of Design)

The biggest fault with our design is the overall shape, the buttons are laid out properly to fit. But the flashiness that comes with the controller's shape also comes with some discomfort when holding and playing. Our design has flair and functionality but comes with minor discomfort.

8.1 Safety

For our design testing day, we had Dr. Marty Kump, the Electrical Computer Engineering 241 Professor and knowledgeable person on PCB controllers assess the safety of our controller. Having years of experience in this field and knowing the applications of this technology rated our controller. We had him rate our controller on a scale from 1-10, 10 being safe for all consumers and 1 being not usable. Where Dr. Kump rated our controller a 10/10 on the scale. To further improve the safety aspect, we could add foam to the reasonably hard and sharp edges of the controller to have no worries for all users.

8.2 Plays Games

The Three games we chose for our controller to play which show the full capabilities are "Snake," "Minesweeper," and "Garden Gnomes." Each game displays a specific function to show the various ways our controller can be used. Snake uses the 4 keycaps which make the game easier than on a keyboard as having buttons in the 4 cardinal directions is more intuitive than arrow keys. Minesweeper showcases the joystick and bumper buttons on the controller which mimic a mouse having with a left and a right click. Garden Gnomes shows reactivity within our controller with minimal delay in the buttons, as the game is a very time sensitive one. All screenshots of these games being completed can be seen above in our "Prototype Analysis." The only tradeoff is the time it takes to become comfortable operating a "mouse" using a controller for "Minesweeper."

8.3 Cost : Bill of Materials

After calculating the unit cost after laying out individual unit prices and scaling. We were able to make sure our design stayed below the price limit.

9. Bill of Materials

Parts List

1. Switches: Gateron Milky Yellow Pro V2 Switches
2. Key Caps: 1U XDA Profile Keycap Blank Crystal Transparent
3. Bumpers: RS PRO PB400OQR1BLK Pushbutton Switch
4. Joystick: Sparkfun 09032
5. Microcontroller: Sparkfun Pro Micro USBC
6. PCBs: Ordered from JLCPCB

R&D BOM (Prototype Cost)

Part #	Part Name	Supplier	Quantity	Bulk Cost	Unit Cost	Shipping Cost	Total
1	Switch	Unikeys	10	\$7	\$0.70	\$0.00	\$7.00
2	Keycap	AliExpress	10	\$3.68	\$0.37	\$0.00	\$3.68
3	Bumper	DigiKey	2	\$4.44	\$2.22	\$6.30	\$10.74
4	Joystick	DigiKey	1	\$5.25	\$5.25	\$6.15	\$11.40
5	Pro Micro	Sparkfun	1	\$7.15	\$7.15	\$7.48	\$14.63
6	PCB	JLCPCB	5	\$8.30	\$1.66	\$2.39	\$10.69

Total Cost	\$58.14
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Table 6 - Research and Development Bill of Materials

Manufacturing BOM (Cost per Unit When Scaling Up)

Part #	Part Name	Supplier	Quantity	Bulk Cost	Unit Cost	Shipping Cost	Total
1	Switch	Amazon	110	\$29.99	\$0.27	\$4.00	\$33.99
		Gateron	105	\$28.35	\$0.27	\$4.30	\$32.65
		Sears	90	\$31.24	\$0.35	\$0.00	\$31.24
2	Key Cap	AliExpress	20	\$5.61	\$0.28	\$0.00	\$5.61
		Amazon	100	\$30.99	\$0.31	\$4.20	\$35.19
		Walmart	20	\$10.84	\$0.54	\$0.00	\$10.84
3	Bumper	Digikey	100	\$165.58	\$1.66	\$6.99	\$172.57
		Newark	50	\$93.00	\$1.86	\$9.99	\$102.99
		RS America	100	\$211.00	\$2.11	\$9.89	\$220.89
4	Joystick	Digikey	5	\$26.25	\$5.25	\$6.12	\$32.37
		Sparkfun	1	\$5.25	\$5.25	\$7.48	\$12.73
		Mouser	1	\$5.25	\$5.25	\$6.10	\$11.35
5	Pro Micro	Sparkfun	1	\$22.50	\$22.50	\$7.48	\$29.98
		AliExpress	1	\$8.32	\$8.32	\$0.00	\$8.32
		Alibaba	1	\$7.36	\$7.36	\$0.00	\$7.36
6	PCB	JLCPCB	100	\$61.20	\$0.61	\$2.39	\$63.59

Table 7 - Manufacturing Bill of Materials

Part #	Orders	Cost
1	4	\$117.70
2	5	\$28.05
3	1	\$172.57
4	100	\$531.10
5	100	\$736.00
6	1	\$63.59
Total Cost		\$1,649.01
Single Cost		\$16.49

Table 8 - 100 Unit Production Cost

We selected the highlighted option as they offered the best price for the most quantity and the lost shipping cost. Some items were harder to buy in bulk than others like the joystick and the pro micro. The difference in the manufacturing cost and the prototype cost comes from buying in bulk. When you order in bulk, the manufacturing companies can cut down on cost as it is cheaper to mass produce a large amount of product than only producing a couple for a prototype. A great example of this is the PCB board. Since it is a custom design, it would be extremely expensive for a company to only make one copy of our design. When they are massed produced, they can make the custom design worth manufacturing with the amount they sell. Just by ordering in bulk, the cost went from \$64.14 per controller to \$16.49 per controller. This is an astronomical difference and shows how important a BOM is when designing and producing a product.

10. Conclusion

The semi-circle game controller design delivers an exceptional balance of functionality, comfort, and affordability. Through multiple tests, we confirmed the controller meets all the criteria required: it is safe, cost-effective, and compact, fitting easily into a standard shipping box. If controllers are ordered in bulk the price of controller went from \$58.14 (under the \$60 limit) to \$16.49 per unit. This makes the controllers very affordable casual users. The performance testing shows the controller effectively can be used in multiple Google Doodle games such as, "Snake," "Minesweeper" and "Garden Gnomes." The average usability score is 7.9/10 after testing. What sets apart this design is its semi-circle shape with 2 bumpers, which allows the user to comfortably hold the controller and offer more buttons to use if games require it. These strengths make the controller practical and allows use for long periods of time without causing pain to the users' hands. While the average comfortability score is 6/10, this offers room for improvements in future models. These improvements could be wireless connection and adding customizable features to allow the user more unique options.

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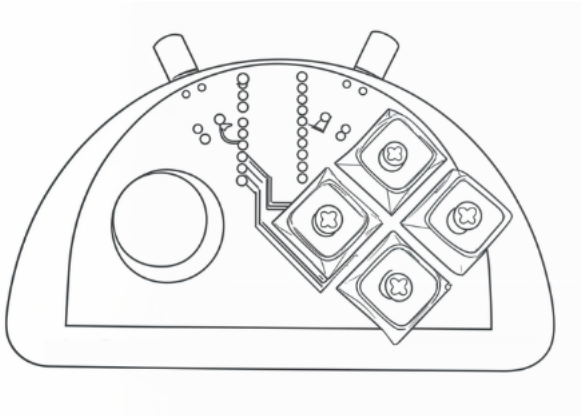
Appendix

- AI was used to create the outline design in Figure 1 (OpenAI 2025)

Controller User Manual

DEN 161 - Group 7

Instructor - Chase Harvey



Jack Meir, Max Requet, Kaiden O'Keefe, Ian Dodson

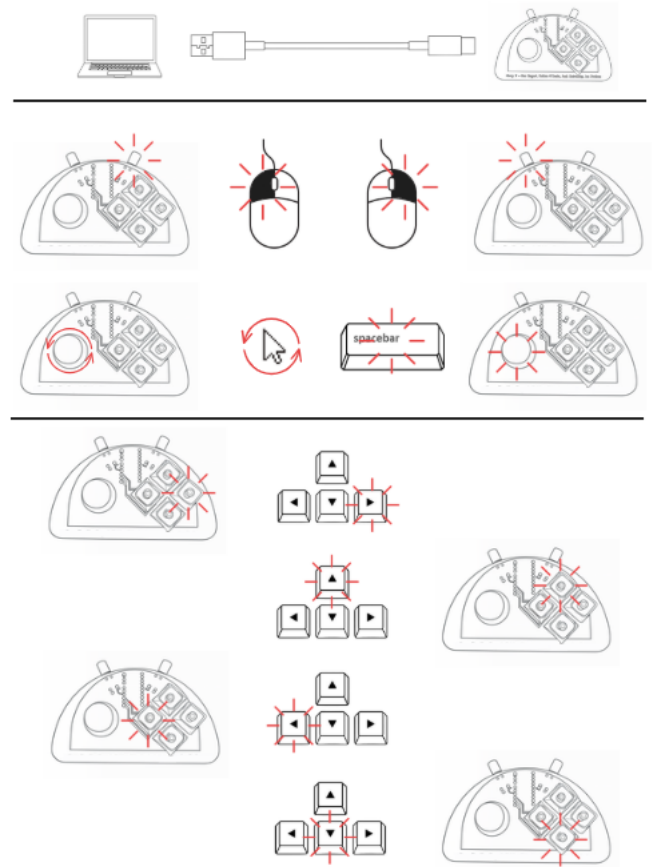


Figure 4 - User Manual